

Project Cheap Adapter: one world, different coordinates

A bounded test of the Platonic Representation Hypothesis, converted into deployable linear adapters and a shared multi-encoder hub.

[Project Documents](#)

EXECUTIVE FINDING

Models do not share raw coordinates or one rigid geometry. They preserve relational structure, and a fitted linear map exposes enough for practical transfer. This is weak, constructive convergence - not a universal representation.

RESEARCH PROGRAM

1

Experiment A

GPT-2 vs Pythia-160M

CKA + layer-wise ridge on 10,000 passages tests whether independent language models share linearly accessible structure.

R² mean 0.737; peak 0.797

2

Experiment B

MobileCLIP to SigLIP 2

One 512x768 ridge matrix aligns phone and server image encoders and is exported as a 0.79 MB fp16 Core ML artifact.

R@5 = 95.3% of native ceiling

3

Cheap Adapter hub

Seven experts, one space

One entry map per encoder reaches a shared 512-d whitened hub, replacing quadratic pairwise translation with linear growth.

N maps, not N(N-1)

EVIDENCE LADDER - STRONGER CLAIMS REQUIRE STRONGER TESTS

1	Shared neighbours	mutual k-NN	strong locally
2	Shared order	distance-rank rho	0.29 cross-modal
3	Shared structure	CKA / RSA	partial
4	Shared subspace	ridge + retrieval	88-97% of ceiling
5	Same geometry	rotation only	rejected: 0.038

The pattern is the result: models agree most on local neighbourhoods, less on global shape, yet become highly interoperable after a fitted linear stretch.

HEADLINE EVIDENCE

0.9183

MLP and linear map tie exactly

0.592

vision adapter held-out R²

0.768

bge-SBERT raw shape rho

96.7%

ConvNeXt hub transfer / native

CLAIM BOUNDARY

Established: useful shared local structure can survive a simple linear change of coordinates on the tested models and data. Not established: identical representations, universal exact linearity, or convergence independent of training data.

From one adapter to the Cheap Adapter hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

SYSTEM DESIGN

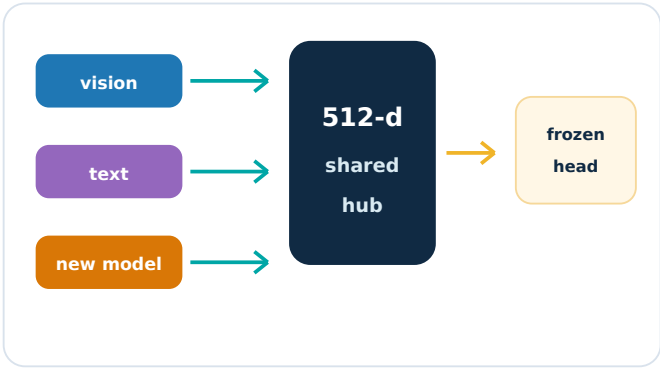
PAIRWISE

7 encoders -> 42 ordered maps

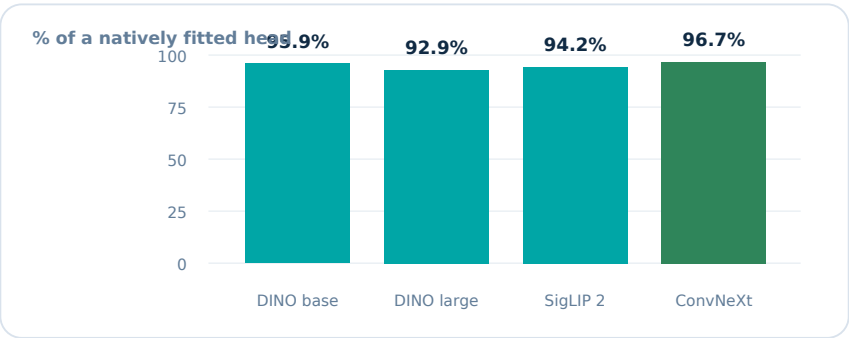
Adding one model requires maps to and from every existing model.

ATLAS HUB

7 encoders -> 7 entry maps



MEASURED TRANSFER - HELD-OUT 1,000-ITEM RETRIEVAL



WHAT TRANSFERRED

- A caption head trained on DINOv2-small serves encoders it never saw.
- SigLIP 2 tests independent lineage; ConvNeXt tests a different architecture.
- A new encoder still needs a fitted entry map: this is adaptation, not zero-shot.

THREE COMPLEMENTARY RESULTS

MAP-FREE

rho = 0.290

Mean cross-modal distance-rank agreement before any map; bge-SBERT reaches 0.768.

COMPOSED

103.6%

Mean hub performance versus bespoke pairwise maps across all 42 ordered pairs.

DEPLOYED

0.79 MB

The fp16 MobileCLIP-to-SigLIP matrix reaches 95.3% of native R@5.

WHY THE INFERENCE IS CREDIBLE

Frozen models only closed-form maps are fitted	Held-out scoring 8,533 train / 1,000 test in the hub	Negative controls shuffle and random maps sit at chance	Nonlinear control MLP ties linear at 0.9183
--	--	---	---

OPERATIONAL INTERPRETATION

Cheap Adapter does not prove that models inhabit one identical latent space. It shows that a compact linear coordinate system can make distinct encoders interoperable without changing their weights, while preserving most task performance on the tested retrieval problems.

Fractal probe: the role of training data

FractalDB-only encoders provide a sharp control: structured training with zero natural photographs, bracketed by matched random-init and ImageNet twins.

THREE TESTS, ONE CONSISTENT READING

G8

Visible geometry

Four map-free metrics reproduce nonzero cross-modal structure. Raw shape rho is 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.

G9

Fractal vs controls

Fractal training adds a small signal above architecture alone, but remains far below natural-image training. Image-vs-text reality dependence is inconclusive.

G10

Functional recovery

A fitted Cheap Adapter entry map plus the frozen caption head finds no FractalDB-specific retrieval advantage over random initialization.

G9 - VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R

0 = random architecture; 1 = ImageNet twin

Fractal CNN

0.09

Fractal ViT

0.21

Matched-norm CNN control: 0.067 fractal vs 0.036 random. The training effect survives preprocessing correction.

Robustness and uncertainty

- CNN image-vs-text R gap: 95% CI [-0.106, +0.100].
- ViT image-vs-text R gap: 95% CI [-0.053, +0.135].
- Corrected drop-1: CNN ordering survives; ViT falls below random.
- CNN-vs-ViT contrast is not interpreted because capacity differs.

G10 - FROZEN HUB/HEAD RETRIEVAL (R@1; OPERATIONAL FLOOR = 0.050)

ENCODER	TRANSFER	NATIVE	RANDOM TWIN	INTERPRETATION
Fractal CNN	0.014	0.014	0.012	at floor; no training-specific gain
Fractal ViT	0.020	0.022	0.017	at floor; no training-specific gain
ImageNet CNN	0.258	0.274	-	clear functional transfer
ImageNet ViT	0.204	0.208	-	clear functional transfer

FINAL CONCLUSIONS

1

Convergence is real but bounded.

The tested models share local relational structure and useful linearly recoverable subspaces, not one exact global representation.

2

Linear maps are functionally sufficient here.

Nonlinear capacity adds no measurable gain; rotation alone fails because anisotropic stretch matters.

3

Natural-world training is the decisive source.

Fractals add a small visible geometric signal, but not a FractalDB-specific semantic transfer advantage through Cheap Adapter.

4

Geometry and utility must be tested separately.

Low raw agreement can coexist with high transfer, while floor-level native retrieval makes transfer ratios meaningless.

SCOPE

Results are bounded to the tested datasets, pooling choices, widths, and linear readouts. Held-out architectures receive fitted entry maps; fractal conclusions cover two released checkpoints; direct CNN-vs-ViT comparison is capacity-confounded.